

MusicPlayer Program Documentation

Overview

The Music Player program is a simple music player application built using Python's Tkinter for the graphical user interface (GUI) and Pygame for audio playback. This program allows users to load a folder containing MP3 files and provides controls to play, pause, and navigate through the songs.

Features

- Load MP3 files from a selected directory.
- Display the list of songs in the directory.
- Play, pause, and navigate through the songs using control buttons.

Class: MusicPlayer

`__init__(self, root)`

Initializes the MusicPlayer instance.

- **Parameters:** ◦ `root`: The Tkinter root window.
- **Description:**
 - Initializes the mixer.
 - Sets up the main window and UI components.
 - Defines global variables for managing the list of songs, the current song, and the paused state.

`setup_ui(self)`

Sets up the user interface components of the music player.

- **Description:**
 - Creates the menu bar with an option to select a folder.
 - Creates a listbox to display the list of songs.
 - Creates control buttons for play, pause, next, and previous actions.
 - Arranges the UI components in the main window.

load_music(self)

Loads music files from the selected directory.

□ Description:

- Opens a dialog to select a directory.
- Filters and loads MP3 files from the selected directory.
- Populates the listbox with the names of the songs.
- Sets the first song as the current song.

play_music(self)

Plays the selected music file or unpauses the playback if paused.

□ Description:

- Checks if there are songs loaded and if playback is not paused.
- Loads and plays the selected song.
- Unpauses the playback if it was paused.

pause_music(self)

Pauses the currently playing music.

□ Description:

- Pauses the music if it is currently playing.
- Sets the paused state to True.

next_music(self)

Plays the next song in the list.

□ Description:

- Selects the next song in the list.
- Clears the previous selection and sets the next song as the current song.
- Plays the next song.

prev_music(self)

Plays the previous song in the list.

□ **Description:**

- Selects the previous song in the list.
- Clears the previous selection and sets the previous song as the current song.
- Plays the previous song.

Usage

1. **Initialize the Music Player:** Create an instance of the `MusicPlayer` class and pass the Tkinter root window to it.

```
python Copy code if
__name__ == "__main__":
    root = Tk()
    app =
    MusicPlayer(root)
    root.mainloop()
```

2. **Load Music:**

- Click on the "Organise" menu and select "Select Folder" to choose a directory containing MP3 files.
- The listbox will display the names of the songs in the selected directory.

3. **Control Playback:**

- Use the play button to start playing the selected song.
- Use the pause button to pause the playback.
- Use the next button to play the next song in the list.
- Use the previous button to play the previous song in the list.

Dependencies

- **Tkinter:** For the graphical user interface.
- **Pygame:** For audio playback.
- **os:** For file and directory operations.